

Radar HRRP Target Recognition Based on Robust Dictionary Learning with Small Training Data Size

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Abstract—In this paper, we develop a robust dictionary learning method for radar high-resolution range profile (HRRP) target recognition, which can utilize the structural similarity between the adjacent HRRPs and overcome the uncertainty of sparse overcomplete representations. Experimental results on the measured HRRP dataset with small training data size show our method can obtain better performance than some other reconstruction algorithms based radar HRRP target recognition methods.

I. INTRODUCTION

A high-resolution range profile (HRRP) is the amplitude of the coherent summations of the complex time returns from target scatterers in each range cell, which represents the projection of the complex returned echoes from the target scattering centers onto the radar line-of-sight. Since it contains abundant target structure signatures, such as target size, scatterer distribution, etc., radar HRRP target recognition has received intensive attention from the radar automatic target recognition (RATR) community [1].

Recent years have witnessed a growing interest in the search for dictionary learning algorithms for sparse signal approximation. Several literatures [2] showed that dictionary learning algorithms is an efficient method to address specific tasks such as natural image processing or data classification.

In radar HRRP target recognition, the method proposed in [3] showed a good recognition performance on measured radar HRRP dataset. The Dictionary Learning (DL) mechanism makes it possible to appropriately share the information among samples from different target-aspects and learn the aspect-dependent dictionary collectively, thus offering the potential to improve the overall recognition performance with small training data size. However, due to the target-aspect sensitivity, amplitudes of adjacent HRRPs may vary somewhat, which would cause the uncertainty of sparse overcomplete representations, and finally affect the recognition performance based on such sparse reconstruction. Hence, it is necessary to propose a radar HRRP target

recognition algorithm that is robust to amplitude variation of adjacent HRRPs.

Let $\mathbf{x}$ be an HRRP sample to be sparsely decomposed over an adaptive overcomplete (redundant) dictionary matrix $\mathbf{D} \in \mathbb{R}^{P \times K}$ that contains K atoms for columns, i.e., a sparse coefficient vector $\mathbf{a}$ has to be found such that $\mathbf{x} = \mathbf{x}_0 + \mathbf{v} = \mathbf{D}\mathbf{a} + \mathbf{v}$, satisfying $\|\mathbf{x} - \mathbf{D}\mathbf{a}\|_p \leq \varepsilon$ (generally, $p = 2$). Donoho, Elad, Temlyakov [4] have shown that if there is a decomposition with $\|\mathbf{a}\|_0 < (1 + M^{-1})/2$, where M denotes the coherence of the dictionary, this decomposition would be stable against $\mathbf{v}$. Therefore, the adaptive overcomplete dictionary should have a property of mutual incoherence, and that the ideal underlying signal has a sufficiently sparse representation, according to a sparsity threshold defined using the mutual coherence of the overcomplete dictionary. In fact, the conditions for stability in [4] are too unduly restrictive to reach; nevertheless we can still design a robust dictionary learning method for a specific task referring to these conditions.

In this paper, we develop a robust dictionary learning method for radar HRRP target recognition. In order to overcome the uncertainty of sparse overcomplete representations caused by amplitude variation of adjacent HRRPs, a robust dictionary learning method is proposed. Experimental comparisons with some other reconstruction algorithms based radar HRRP target recognition methods, such as PCA-based minimum reconstruction error approximation, K-SVD algorithm based radar HRRP recognition [3] etc., demonstrate the better recognition performance on the measured radar HRRP dataset with small training data size.

II. REVIEW OF DICTIONARY LEARNING MODEL

A. Dictionary Learning Model

Consider a linear model $\mathbf{D}\mathbf{a} = \mathbf{x}$, where $\mathbf{x} \in \mathbb{R}^P$ and $\mathbf{D} \in \mathbb{R}^{P \times K}$ ($\mathbf{D}$ is a full-rank matrix as $P < K$). An infinite

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number of solutions are available for this model, but the solution $\mathbf{a}$ with the fewest number of nonzero coefficients is certainly an appealing representation.

There are two stages for the implementation of dictionary learning model.

1) *Sparse coding*: With the given dictionary $\mathbf{D} \in \mathbb{R}^{P \times K}$ and signal $\mathbf{x} \in \mathbb{R}^P$, we can get the sparse representation $\mathbf{a} \in \mathbb{R}^K$ by solving:

$$\min_{\mathbf{a}} \|\mathbf{a}\|_0 \quad \text{s.t.} \quad \|\mathbf{x} - \mathbf{D}\mathbf{a}\|_2^2 \leq \varepsilon \quad (1)$$

$$\text{or} \quad \min_{\mathbf{a}} \|\mathbf{x} - \mathbf{D}\mathbf{a}\|_2^2 \quad \text{s.t.} \quad \|\mathbf{a}\|_0 \leq T \quad (2)$$

where $\|\mathbf{a}\|_0$ represents the l_0 norm, counting the number of non-zero elements in $\mathbf{a}$. ε and T represent the noise variance and sparsity level, respectively. However, the exact solution of (1) or (2) requires searching over all combinations of non-zero components, which is a NP-hard problem. There are two ways to get an approximate solution. One approach is to relax the l_0 constraint to a l_1 constraint, yielding a convex problem, which is usually more precise but less efficient and can be solved by basis pursuit (BP). An alternative way is using greedy algorithms, such as matching pursuit (MP), orthogonal matching pursuit (OMP) [3].

2) *Dictionary update*: Designing dictionaries to better fit the linear model $\mathbf{D}\mathbf{a} = \mathbf{x}$ can be done by either selecting one from a prespecified set or adapting the dictionary to a set of training signals. A prespecified dictionary is appealing due to its simple computations. But using dictionaries adapted to the data, such as K-SVD and method of optimal directions (MOD), might better fit the signal of interest, leading to improved performance over a prespecified dictionary.

B. Stable Recovery of Sparse Overcomplete Representations

Consider a linear model $\mathbf{x} = \mathbf{x}_0 + \mathbf{v} = \mathbf{D}\mathbf{a} + \mathbf{v}$, where $\mathbf{x}_0 \in \mathbb{R}^P$ is an ideal noiseless signal. Donoho, Elad, Temlyakov [4] have proved that when sufficient sparsity is present, i.e., $\|\mathbf{a}\|_0 < (1 + M^{-1})/2$, solving (1) enables stable recovery.

M denotes the mutual coherence of the dictionary

$$M = M(\mathbf{D}) = \max_{1 \leq k, j \leq m, k \neq j} |\mathbf{G}(k, j)| \quad (3)$$

where Gram matrix $\mathbf{G} = \mathbf{D}^T \mathbf{D}$.

On the other hand, it is known that a sparse decomposition with $\|\mathbf{a}\|_0 < (1/2)\text{spark}(\mathbf{D})$ is unique, but its stability against $\mathbf{v}$ has been proved only for highly more restrictive decompositions satisfying $\|\mathbf{a}\|_0 < (1 + M^{-1})/2$, because usually $(1 + M^{-1})/2 \ll (1/2)\text{spark}(\mathbf{D})$ [5]. Therefore, it is clear that tighter bound $\|\mathbf{a}\|_0 < (1 + M^{-1})/2$ can guarantee to get the ‘Correct’ support with (1). However, the conditions for stability are unduly restrictive and ‘excessive pessimism’. They used worst case reasoning exclusively, deriving

conditions which must apply to every dictionary, every sparse representation and every bounded noise vector [5].

III. RADAR HRRP TARGET RECOGNITION BASED ON ROBUST DICTIONARY LEARNING

A. The preprocessing of the measured HRRP dataset

In the target recognition based on HRRP, a challenging task is how to deal with the time-shift and amplitude-scale sensitivity of HRRP.

1) *Time-Shift Sensitivity*: In order to decrease the computation complexity, an HRRP is only a part of received radar echo extracted by a range window, in which target signal is included. Thus the position of the target signal in an HRRP can vary with the measurement. There are many useful feature extraction methods, such as the spectrum amplitude of real HRRPs, power spectrum and etc., to deal with the time-shift sensitivity of HRRP. With the given HRRP $x(t)$, the Fourier transform of the $x(t - \tau)$ can be represented as:

$$X'(w) = X(w) \exp(-jw\tau) \quad (4)$$

$$|X'(w)|^2 = |X(w)|^2 \quad (5)$$

Obviously, the power spectrum is a feature extraction method invariant with the time-shift for HRRPs. Therefore, we take the power spectrum as the feature in our method. Because the power spectrum has a symmetrical structure, half of the power spectrum sequence can be used as the input signal.

2) *Amplitude-Scale Sensitivity*: The amplitude-scale sensitivity of HRRP comes from the fact that the intensity of an HRRP is a function of radar transmitting power, target distance, radar antenna gain, radar receiver gain, radar system losses and so on. Thus HRRPs measured by different radars or under different conditions will have different amplitude-scales. Here we take amplitude normalization to deal with the amplitude-scale sensitivity of HRRPs.

B. Conditions for amplitude variation stability

Because HRRPs are continuous and change slowly, there exist large correlation coefficients among the adjacent HRRPs from a target-aspect sector without the scatterers’ motion through range cells (MTRC) [1, 3]. Hence, an intuitive idea is that the sparse representations of adjacent HRRPs should have similar supports and lower variance. Due to the target-aspect sensitivity, amplitudes of adjacent HRRPs may vary somewhat, which would cause the uncertainty of sparse overcomplete representations, and finally affect the recognition performance based on such sparse reconstruction. Similarly, there also exists amplitude variation among the power spectra of adjacent HRRPs. Therefore, it is necessary to propose a radar HRRP target recognition algorithm that is robust to the variation among the adjacent HRRPs. The conditions for variation stability are as follows, referred to the conclusions drawn in [4]:

1) *Ideal noiseless*: There are two ideal noiseless adjacent samples $\mathbf{x}_1$, $\mathbf{x}_2$ with the ‘sufficient’ sparse representations

$\mathbf{x}_1 = \mathbf{D}\mathbf{a}_1$ and $\mathbf{x}_2 = \mathbf{D}\mathbf{a}_2$ respectively. The assumption here is that the sparse representation $\mathbf{a}_1$ and $\mathbf{a}_2$ have at most N nonzeros.

Consider a linear model $\mathbf{x}_2 = \mathbf{D}\mathbf{a}_2 = \mathbf{x}_1 + \mathbf{v} = \mathbf{D}\mathbf{a}_1 + \mathbf{v}$. When $\|\mathbf{v}\|_2$ (the energy of amplitude variation) satisfies some conditions [4], and $\|\mathbf{a}_1\|_0, \|\mathbf{a}_2\|_0 < (1 + M^{-1})/2$, then $\mathbf{a}_1, \mathbf{a}_2$ are the unique sparsest such representations of $\mathbf{x}_1, \mathbf{x}_2$ respectively. The sparse representations $\mathbf{a}_1$ and $\mathbf{a}_2$ obey:

$$\text{support}(\mathbf{a}_1) = \text{support}(\mathbf{a}_2) \quad (6)$$

$$r(\mathbf{a}_1, \mathbf{a}_2) = \|\mathbf{a}_2 - \mathbf{a}_1\|_2^2 \leq \varepsilon_1 = \frac{4\|\mathbf{v}\|_2^2}{1 - M(2N - 1)} \quad (7)$$

where $\text{support}(\mathbf{a})$ denotes the position of non-zero elements in $\mathbf{a}$.

2) *Presence of Noise*: Suppose the existence of adjacent noiseless samples $\mathbf{x}_{1,0} = \mathbf{D}\mathbf{a}_{1,0}$, $\mathbf{x}_{2,0} = \mathbf{D}\mathbf{a}_{2,0}$ and noisy observations $\mathbf{x}_1 = \mathbf{x}_{1,0} + \mathbf{n}_1$, $\mathbf{x}_2 = \mathbf{x}_{2,0} + \mathbf{n}_2$. $\mathbf{a}_1$ and $\mathbf{a}_2$ are the sparse representations of the noisy signals $\mathbf{x}_1, \mathbf{x}_2$ respectively by solving (1).

Consider the linear model $\mathbf{x}_{2,0} = \mathbf{x}_{1,0} + \mathbf{v}$. When $\|\mathbf{v}\|_2$, $\|\mathbf{n}_1\|_2$ and $\|\mathbf{n}_2\|_2$ satisfy some conditions [4], and $\|\mathbf{a}_{1,0}\|_0, \|\mathbf{a}_{2,0}\|_0 < (1 + M^{-1})/2$, then $\mathbf{a}_1, \mathbf{a}_2$ are the unique sparsest such representations of $\mathbf{x}_1, \mathbf{x}_2$ respectively. The sparse representations $\mathbf{a}_1$ and $\mathbf{a}_2$ obey:

$$\text{support}(\mathbf{a}_1) = \text{support}(\mathbf{a}_2) \quad (8)$$

$$r(\mathbf{a}_1, \mathbf{a}_2) = \|\mathbf{a}_2 - \mathbf{a}_1\|_2^2 \leq \frac{4(\|\mathbf{v}\|_2^2 + \|\mathbf{n}_1\|_2^2 + \|\mathbf{n}_2\|_2^2)}{1 - M(2N - 1)} \quad (9)$$

Actually, it is clear that the conditions for amplitude variation stability are too unduly restrictive to reach. Alternatively, we develop a robust dictionary learning method for radar HRRP target recognition with constraints on sparse representations.

C. Robust HRRP Recognition Method

1) *Dictionary training Stage*: Training samples $\mathbf{X}_l \in \mathbb{R}^{P \times N_l}$ ($l = 1, 2, \dots, L$, l denotes class label) are defined as the power spectra of real HRRPs with amplitude normalization. Set the initial dictionary $\mathbf{D}_l \in \mathbb{R}^{P \times K}$ with l_2 normalized columns. Learning an adaptive overcomplete dictionary $\mathbf{D}_l^*$ with a fixed number K of atoms, is addressed by solving the following minimization problem.

$$\begin{aligned} \min_{\mathbf{D}_l, \{\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_{N_l}\}} & \left(\sum_{i=1}^{N_l} \|\mathbf{x}_i - \mathbf{D}_l \mathbf{a}_i\|_2^2 \right) + \lambda \frac{1}{N_l} \sum_{i=1}^{N_l} \|\mathbf{a}_i - \mathbf{m}\|_2^2 \\ \text{s.t.} \quad & \mathbf{m} = \frac{1}{N} \sum_{i=1}^{N_l} \mathbf{a}_i \\ & \|\mathbf{a}_i\|_0 \leq C \\ & \text{support}(\mathbf{a}_i) = \text{support}(\mathbf{a}_j), \forall i, j \in \{1, 2, \dots, N_l\} \end{aligned} \quad (10)$$

where the parameter $\lambda \geq 0$ controls the trade-off between reconstruction and penalization.

A greedy algorithm is used for the sparse coding, which has been shown to be very efficient. And a least-squares problem is solved for updating all atoms. K-SVD [6] is also an alternative way.

2) *HRRP target Recognition stage*: Denote the normalized power spectrum of a test HRRP sample as $\mathbf{y}$. Thus, the class label l^* for the test HRRP sample $\mathbf{y}$ can be estimated as:

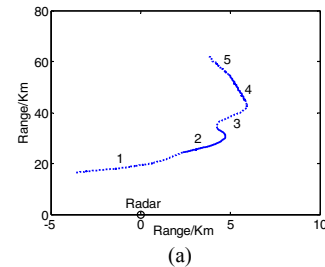
$$l^* = \arg \min_l \|\mathbf{y} - \mathbf{D}_l^* \mathbf{a}_l\|_2 \quad (11)$$

IV. EXPERIMENTAL RESULTS

We carried out several experiments on HRRP data, trying to show the feasibility of the proposed algorithm. The parameters of the targets and radar are shown in Table 1, and the projections of target trajectories onto the ground plane are shown in Fig. 1, from which the aspect angle of an airplane can be estimated according to its relative position to radar. The original training data in our experiments are segmented, with 35 aspect-frames for Yak-42, 50 for Cessna Citation S/II, and 50 for An-26, of which each frame has 21 HRRPs (small training data size) [1,3,7]. In addition, the measured HRRP is a 256-dimensional vector.

TABLE I. PARAMETERS OF PLANES AND RADAR IN THE ISAR EXPERIMENT

radar parameters	Center frequency		5520 MHz
	bandwidth		400 MHz
planes	length (m)	width (m)	height (m)
Yark-42	36.38	34.88	9.83
An-26	23.80	29.20	9.83
Cessna Citation S/II	14.40	15.90	4.57



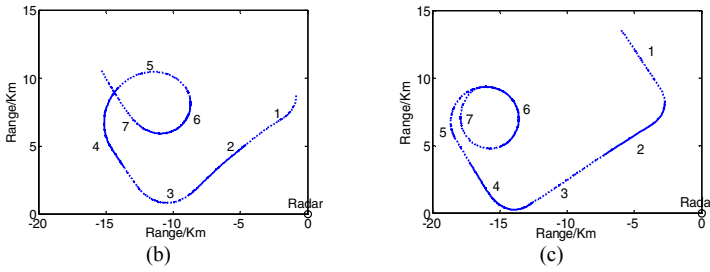


Figure 1. Projections of target trajectories onto the ground plane: (a) Yark-42; (b) An-26; (c) Cessna Citation S/II.

TABLE II. THE CONFUSION MATRICES AND AVERAGE RECOGNITION RATES WITH SMALL TRAINING DATA SIZE (%)

	PCA model			AGC model		
	Yark-42	Cessna Citation S/II	An-26	Yark-42	Cessna Citation S/II	An-26
Yark-42	85.00	9.67	5.33	89.67	8.67	1.67
Cessna Citation S/II	0.22	75.78	24.00	0.22	76.44	23.33
An-26	2.22	7.33	90.44	6.00	11.56	82.44
Average recognition rates P_{cc} (%)	83.74			82.58		

	K-SVD model			Robust DL model		
	Yark-42	Cessna Citation S/II	An-26	Yark-42	Cessna Citation S/II	An-26
Yark-42	98.67	0	1.33	96.33	1.00	2.67
Cessna Citation S/II	1.78	84.00	14.22	0	92.44	7.56
An-26	13.11	6.67	80.22	3.11	9.11	87.78
Average recognition rates P_{cc} (%)	87.63			92.19		

Table 2 shows that the proposed robust DL algorithm based radar HRRP recognition method has better recognition rate than some other some other reconstruction algorithms based radar HRRP target recognition methods with a small training data size, such as PCA-based minimum reconstruction error approximation, K-SVD algorithm based radar HRRP recognition [3]. AGC model [7] is also brought into comparison here in order to validate our method. Furthermore, the recognition accuracies versus the size of training dataset are shown in Fig.2.

Overcomplete dictionary based sparse representation is a more efficient and reasonable way for data description. The DL mechanism makes it possible to appropriately share the information among samples from different target-aspects and

learn the aspect-dependent dictionary collectively, thus offering the potential to improve the overall recognition performance with small training data size. Hence, K-SVD model and Robust DL model have a better recognition performance than PCA model or AGC model. The robust DL algorithm has a better recognition performance than K-SVD method [3], because the proposed method not only consider the structural similarity between the adjacent HRRPs, but also overcome the uncertainty of sparse overcomplete representations.

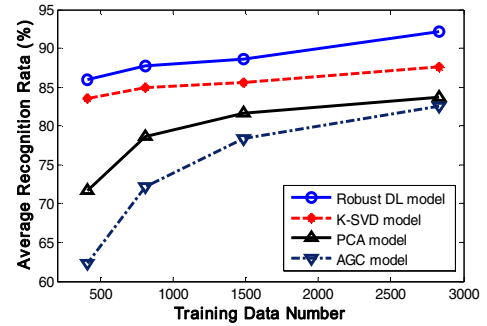


Figure 2. Variation of the recognition performance with training data size

V. CONCLUSION

The robust dictionary learning method for radar HRRP target recognition proposed in this paper not only consider the structural similarity between the adjacent HRRPs, but also overcome the uncertainty of sparse overcomplete representations. Experimental results indicate that the recognition performance of our method is likely to be superior to some other existing radar HRRP target recognition algorithms based on reconstruction model with a small training data size.

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